

i. What is Science?

- **What is Science?:** Science is a systematic enterprise that builds and organizes knowledge in the form of testable explanations and predictions about the universe. It involves observation, experimentation, and the formulation of theories to understand natural phenomena. Key characteristics of science include empirical evidence, repeatability, and peer review.

ii. What is the Scientific Method?

- **Scientific Method:** The scientific method is a systematic process used to investigate phenomena, acquire new knowledge, or correct and integrate previous knowledge. It involves several steps:
 - **Observation:** Gathering data and noticing patterns.
 - **Hypothesis Formation:** Proposing a tentative explanation based on observations.
 - **Experimentation:** Designing and conducting experiments to test the hypothesis.
 - **Analysis:** Interpreting the results of the experiments.
 - **Conclusion:** Determining whether the hypothesis is supported or refuted by the results.
 - **Replication:** Repeating experiments to verify results.
 - **Theory Formation:** Developing a coherent explanation that integrates multiple hypotheses and observations.

iii. The Problem of Induction

- **The Problem of Induction:** Introduced by David Hume, the problem of induction questions the justification for the inference of general laws from specific observations. Induction involves drawing general conclusions based on finite sets of observations, but there is no logical certainty that future observations will conform to past patterns.
 - **Example:** Observing that the sun rises every day does not logically guarantee that it will rise tomorrow.
 - **Hume's Challenge:** Since inductive reasoning relies on the assumption that the future will resemble the past, it cannot be logically justified purely by empirical evidence.

iv. Karl Popper's Theory of Falsificationism

- **Karl Popper's Theory of Falsificationism:** Popper proposed falsificationism as a solution to the problem of induction. According to Popper, scientific theories cannot

be conclusively proven but can be falsified. A theory is scientific if it makes bold predictions that can be tested and potentially refuted by empirical evidence.

- **Key Concepts:**

- **Demarcation Criterion:** The criterion distinguishing scientific theories from non-scientific ones is their falsifiability.
- **Bold Conjectures and Refutations:** Scientific progress occurs through bold conjectures that are rigorously tested and potentially refuted, leading to the development of better theories.
- **Popper's View:** Science advances by eliminating false theories and refining the surviving ones.

v. Thomas Kuhn's Theory of Scientific Revolutions

- **Thomas Kuhn's Theory of Scientific Revolutions:** Kuhn introduced the concept of paradigm shifts in his book "The Structure of Scientific Revolutions." He argued that scientific progress is not a gradual accumulation of knowledge but rather a series of revolutionary changes in which one paradigm is replaced by another.

- **Key Concepts:**

- **Paradigm:** A framework of theories, methods, and standards that defines legitimate research within a scientific community.
- **Normal Science:** Periods of research conducted under the prevailing paradigm, where scientists solve puzzles within the framework of the existing paradigm.
- **Anomalies:** Observations or problems that cannot be explained by the current paradigm.
- **Crisis and Revolution:** When anomalies accumulate, they may lead to a crisis and the eventual replacement of the old paradigm with a new one.
- **Paradigm Shift:** A fundamental change in the basic concepts and experimental practices of a scientific discipline.

The philosophy of science addresses crucial questions about the nature and methodology of scientific inquiry. It examines the scientific method, the problem of induction, and the development of scientific knowledge through theories like Popper's falsificationism and Kuhn's paradigm shifts. By exploring these concepts, we gain a deeper understanding of how science operates, evolves, and contributes to our knowledge of the world.